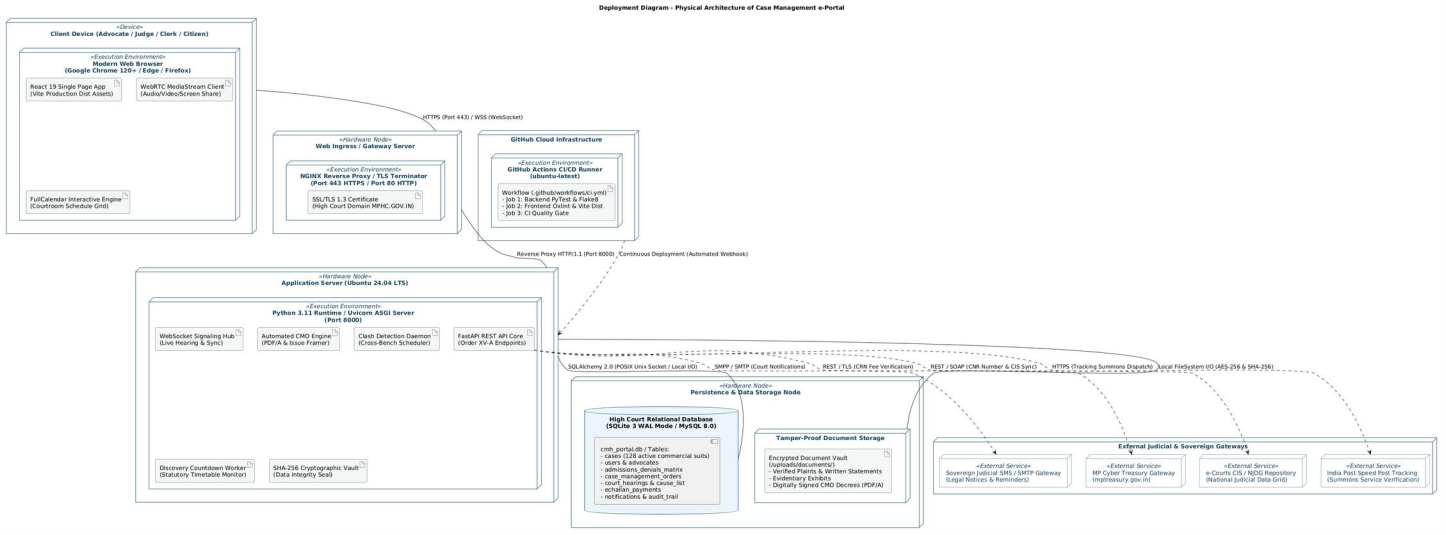
5. UML Deployment Diagram

Hardware Node Topology, Execution Environments, and Secure Protocols



ARCHITECTURAL & STATUTORY CPC HIGHLIGHTS:

- Client Devices: Browser execution environment running React 19 SPA over TLS 1.3 HTTPS and WSS protocols.
- Web Ingress Node: NGINX reverse proxy providing SSL termination and protective DDoS rate limiting.
- Application Server Node: Ubuntu 24.04 LTS host running Python 3.11 with Uvicorn ASGI on port 8000.
- Storage & External Nodes: POSIX file vault for PDF/A exhibits, connecting to MP Treasury and India Post APIs.